



AIMS05-01-001

Issue 5

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July 2013

AIMS
Airbus Material Specification
Carbon Fibre Reinforced Epoxy Prepreg
Unidirectional tape / 180°C curing
Standard modulus fibre
Material Specification

Published and distributed by :
AIRBUS S.A.S.
ENGINEERING DIRECTORATE
31707 BLAGNAC Cedex
FRANCE

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1 Scope

This Material Specification defines the requirements for a group of materials each consisting of a carbon fibre reinforced unidirectional tape prepreg, curing at 180°C for aerospace application. In addition, this Material Specification defines the requirements for the resin film, curing at 180°C for aerospace application, to be used with the corresponding prepreg (the one with the same resin system).

The requirements specified in this Material Specification shall not be used as stress allowables.

Unless otherwise specified by the drawing, the order or the inspection schedule, the present specification shall be used in conjunction with the referenced documents.

2 Application

This material is used for manufacturing primary airframe and aerospace structures (sandwich and monolithic).

Materials to be delivered according to ABS 5139.

The prepreg shall have an operating temperature range of –55°C to 95°C.
Nominal cured ply thickness: to be found in ABS 5139

The resin film (whichever the areal weight is, i.e.: 250 g/m²) is to be used with the corresponding prepreg only, and for local application only.
Restrictions concerning the application, if applicable

3 Normative References

This Airbus specification incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this Airbus specification only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

AIMS 05-01-000 – Part 1	Airbus Material Specification; Carbon fibre reinforced epoxy prepreg, General requirements – Technical Specification
AIMS 05-01-000 – Part 2	Airbus Material Specification; Carbon fibre reinforced epoxy prepreg Qualification and batch release testing. UD tape/180 °C curing class –Technical Specification
ABS 5139	Airbus Standard; Carbon fibre reinforced epoxy prepreg – unidirectional tape/180 °C-curing class, standard modulus fibre - Structural material
AIMS05-01-000 Part 100	Carbon fiber reinforced epoxy prepreg Structural material Acceptance Testing and Storage Life Prolongation

4 Technical Requirements

4.1 General

Specification values are minimum except where stated.

4.1.1. Material description

See Table 1

4.1.2. Worklife and storage life

See Table 2

4.1.3. Processing

The laminates or test purposes shall be cured in an autoclave. The curing cycle shall comply with the general requirements reported in Table 3.

The curing cycle shall be agreed between the manufacturer and Airbus at the onset of the material qualification programme and shall be used to produce all test specimens for qualification testing. The agreed curing cycle shall be specified in the relevant IPS and shall be used for the manufacture of all the batch release test specimens.

4.2. Specific (grade A, B, C)

4.2.1. Physical-chemical properties

Uncured and cured properties see Table 4.

4.2.2. Mechanical properties

See Tables 5, 6 and 7.

4.3. Additional requirements for the resin film (grade 0)

4.3.1. Physical-chemical properties

Uncured and cured properties see table 8.

4.3.2. Mechanical properties

See table 9.

5 Designation

Refer to ABS 5139.

6 Technical Specification

The general technical requirements and responsibilities corresponding to this Material Specification are described in the Technical Specification AIMS 05-01-000 Part 1 and Part 2.

7 Incoming inspection

Materials qualified against this AIMS shall undergo "Technical Incoming Inspection" as per AIMS05-01-000 Part 100 prior to use.

Table 1: – Material description

Characteristic	Description			
Form	Prepreg			Resin film
Method of production	Cured in an autoclave at 180°C			
Material type	UD carbon fibre tape/epoxy resin			epoxy resin
Formulation	Fibre: Carbon, standard modulus Resin: Epoxy, toughened, controlled flow			toughened epoxy controlled flow
Resin content by weight, uncured	Grade A ($m_F = 134 \text{ g/m}^2$) 35 % nominal or 40 % nominal	Grade B ($m_F = 196 \text{ g/m}^2$) 34 % nominal	Grade C ($m_F = 268 \text{ g/m}^2$) 35 % nominal or 40 % nominal	Grade 0 Resin film areal weight: see relevant IPS
Service temperature range	-55°C up to 95°C			

Table 2: Worklife and storage conditions

No.	Properties	Shop conditions	Unit	Requirements
1	Storage life $5 \pm 2^{\circ}\text{C}$	-	months	-
2	Storage life -18°C ¹⁾	-	months	≥ 12
3	Worklife ²⁾ (tack life)	$30^{\circ}\text{C}/75\% \text{ RH}$	days	≥ 10
		$25 \pm 2^{\circ}\text{C} / 45 \pm 10\% \text{ RH}$	days	≥ 10
¹⁾ Deviations: see relevant IPS				
²⁾ The prepreg can be exposed an additional minimum of 5 days at the above mentioned shop conditions, provided that the lay-up is fully bagged and ready for use				

Table 3: Processing conditions

Cure type	Autoclave		
Cure conditions ¹⁾	Heating rate	°C/min	0,2 – 5
	Cure temperature	°C	180 +10/-5
	Cure time	min	120 - 240
	Cure pressure	MPa	0,3 – 1,1
	Cooling rate	°C/min	0,2 – 3,5
	Vacuum (abs)	MPa	0,005 – 0,1
¹⁾ Deviations: see relevant IPS			

Table 4: Physical-chemical properties (grade A, B, and C)

No.	Properties	Test methods	Units	Requirements				
Material code				AIMS05-01-001				
				A	B	C	D	E
Uncured				Grade A		Grade B	Grade C	
1	Prepreg mass/unit area	EN 2557	g/m ²	206 ± 12	223 ± 14	293 ± 20	412 ± 24	446 ± 28
2	Fibre mass/unit area	EN 2559	g/m ²	134 ± 6		196 ± 5	268 ± 12	
3	Resin density	ISO 1183 Method A	g/cm ³	see relevant IPS				
4	Fibre density	ISO 10119	g/cm ³	1,78 ± 0,04				
5	Volatile content	EN 2558	%	≤ 2				
6	Resin content	EN 2559	%	35 ± 2,5	40 ± 2,5	34 ± 2	35 ± 2,5	40 ± 2,5
7	Physic-chemical tests	AIMS 05-01-000 part 1	-	see relevant IPS				
8	Resin flow	EN 2560	%	> 2	> 6	see relev. IPS	> 2	> 6
9	Tack	To be agreed between manufacturer and Airbus National Company	-	see relevant IPS				
Cured								
10	Coefficient of thermal expansion	ISO 11359-2	-	see relevant IPS				
11	Water uptake	EN 3615, 70°C/85% RH	%	≤ 1,4				
12	Water uptake	EN 2378, 70°C/water	%	see relevant IPS				
13	Extent of cure	AITM 3.0008	%	see relevant IPS				
Note: The stated requirements are average values								

Table 5: Mechanical properties – dry (grade A, B, C)

No	Properties	Test Methods	Units	Requirements							
				Dry							
				-55°C		RT		70°C		120°C	
				$\bar{X}$	Xmin	$\bar{X}$	Xmin	$\bar{X}$	Xmin	$\bar{X}$	Xmin
1	Glass transition temperature, Tg onset	AITM 1.0003	(°C)	Xmin = 150							
2a	Tensile strength 0°	EN 2561 type B	(MPa)	1800	1650	1800	1650	1780	1630	-	-
2b	Tensile modulus 0°	EN 2561 type B	(GPa)	135 ± 15		135 ± 15		135 ± 15		-	
2c	Poissons Ratio 0/90°	EN 2561 type B	-	-		-		-		-	
3a	Tensile strength 90°	EN 2597 type B ¹⁾	(MPa)	55	50	60	50	-	-	-	-
3b	Tensile modulus 90°	EN 2597 type B ¹⁾	(GPa)	10 ± 1,7		10 ± 1,7		-		-	
4a	Compression strength 0°	EN 2850 type A ²⁾	(MPa)	1330	1210	1200	1135	1100	1000	-	-
4b	Compression modulus 0°	EN 2850 type A ²⁾	(GPa)	125 ± 15		125 ± 15		125 ± 15		-	
5a	Compression strength 90°	EN 2850 type B	(MPa)	-	-	200	170	-	-	-	-
5b	Compression modulus 90°	EN 2850 type B	(GPa)	-		10 ± 1,5		-		-	
6a	In plane shear strength ⁴⁾ Grade A + B Grade C	AITM 1.0002	(MPa)	90 70	85 65	90 70	85 65	85 65	75 60	80 ³⁾ 65	70 ³⁾ 55
6b	In plane shear modulus	AITM 1.0002	(GPa)	5,0 ± 1,5		5,0 ± 1,5		4,8 ± 1,5		4,5 ± 1,5	
7	ILSS 0°	EN 2563	(MPa)	135	123	100	95	72	65	63	58
8	Compression after impact ^{4) 5)} Grade A Grade B + C	AITM 1.0010	(MPa)	-	-	220 160	-	-	-	-	-
9	G _{1C}	AITM 1.0005	(J/m ²)	-	-	225	215	-	-	-	-
10	G _{2C}	AITM 1.0006	(J/m ²)	-	-	640	580	-	-	-	-

¹⁾ or type A if required

²⁾ or type B if required

³⁾ Guide values, statistically not secured

⁴⁾ Grades definition see Table 1

⁵⁾ Impact Energy 6,7 J/mm

Note:

- X means minimum average
- X_{min} means minimum individual value.
- All requirements for modulus are related to mean values only
- Determination of Inplane shear modulus with tangent in origin of coordinates
- Properties No. 2a, 2b, 4a, and 4b calculated to a fibre volume content of 60 %

Table 6: Mechanical properties – hot/wet (grade A, B, C)

No.	Properties	Test Method	Units	Requirements							
				Wet 70°C/85% R.H.							
				RT		70°C ³⁾		90°C		120°C	
				$\bar{X}$	Xmin	$\bar{X}$	Xmin	$\bar{X}$	Xmin	$\bar{X}$	Xmin
1	Glass transition temperature, Tg onset	AITM 1.0003	(°C)	X min = 120							
2a	Tensile strength 0°	EN 2561 type B	(MPa)	-	-	1670	1450	1670	1450	-	-
2b	Tensile modulus 0°	EN 2561 type B	(GPa)	-		135 ± 15		135 ± 15		-	
2c	Poissons Ratio 0/90°	EN 2561 type B	-	-		-		-		-	
3a	Tensile strength 90°	EN 2597 type B ¹⁾	(MPa)	-	-	30	25	30	25	-	-
3b	Tensile modulus 90°	EN 2597 type B ¹⁾	(GPa)	-		9,5 ± 2		7,7 ± 1,7		-	
4a	Compression strength 0°	EN 2850 type A ²⁾	(MPa)	-	-	1050	990	900	800	-	-
4b	Compression modulus 0°	EN 2850 type A ²⁾	(GPa)	-		125 ± 15		125 ± 15		-	
5a	Compression strength 90°	EN 2850 type B	(MPa)	-	-	165	160	120	110	-	-
5b	Compression modulus 90°	EN 2850 type B	(GPa)	-		8,25 ± 1,75		8,25 ± 1,75		-	
6a	In plane shear strength ⁴⁾ Grade A + B Grade C	AITM 1.0002	(MPa)	-	-	80 65	70 55	75 60	65 50	-	-
6b	In plane shear modulus	AITM 1.0002	(GPa)	-		4,8 ± 1,5		4,5 ± 1,5		-	
7	ILSS 0°	EN 2563	(MPa)	-	-	68	63	57	41	-	-
8	Compression after impact ^{4) 5)} Grade A Grade B + C	AITM 1.0010	(MPa)	-	-	-	-	-	-	-	-
9	G _{1C}	AITM 1.0005	(J/m ²)	-	-	-	-	-	-	-	-
10	G _{2C}	AITM 1.0006	(J/m ²)	-	-	-	-	-	-	-	-

¹⁾ or type A if required

²⁾ or type B if required

³⁾ Guide values, statistically not secured

⁴⁾ Grades definition see Table 1

⁵⁾ Impact Energy 6,7 J/mm

Note:

- X means minimum average
- X_{min} means minimum individual value.
- All requirements for modulus are related to mean values only
- Determination of Inplane shear modulus with tangent in origin of coordinates
- Properties No. 2a, 2b, 4a, and 4b calculated to a fibre volume content of 60 %

Table 7: Mechanical properties / resistance to aggressive fluids (grades A, B, C)

No	Properties	Test Method	Units	Requirements											
				Dry								Wet 70°C/85% R.H.			
				-55°C		RT		90°C		120°C		70°C		90°C	
				$\bar{X}$	Xmin	$\bar{X}$	Xmin	$\bar{X}$	Xmin	$\bar{X}$	Xmin	$\bar{X}$	Xmin	$\bar{X}$	Xmin
1a	ILSS (reference)	EN 2563	(MPa)	-	-	80	72	-	-	-	-	-	-	42	35
1b	ILSS skydrol, 70°C/1000 h	EN 2563	(MPa)	-	-	-	-	(1)	(1)	-	-	-	-	-	-
1c	ILSS skydrol/water, 50°C/1000 h	EN 2563	(MPa)	-	-	-	-	(1)	(1)	-	-	-	-	-	-
1d	ILSS fuel, RT/1000 h	EN 2563	(MPa)	-	-	-	-	(1)	(1)	-	-	-	-	-	-
1e	ILSS MEK, RT/1 h	EN 2563	(MPa)	-	-	(1)	(1)	-	-	-	-	-	-	-	-
2a	In plane shear strength (phenolic paint stripper)	AITM 1.0001	(MPa)	-	-	(1)	(1)	-	-	-	-	-	-	-	-
2b	In plane shear strength (non-phenolic paint stripper)	AITM 1.0001	(MPa)	-	-	(1)	(1)	-	-	-	-	-	-	-	-

¹⁾ Values after conditioning will not have a decrease higher than the corresponding one in comparison with the wet (70°C/85%R.H.) reference value at the equivalent temperature.

Note: • ILSS: interlaminar shear strength
 • Lay-up sequence for interlaminar shear strength acc. To AIMS 05-01-000 Part 1
 • Peel ply to be used for surface preparation
 • X means minimum average value.
 • X_{min} means minimum individual value

Table 8: Physical-chemical properties for the resin film (grade 0)

No.	Properties	Test methods	Unit	Requirements
Uncured				Grade 0
1	Resin mass/unit area	EN 2557	g/m ²	See relevant IPS
2	Resin density	ISO 1183 Method A	g/cm ³	See relevant IPS
3	Volatile content	EN 2558	%	≤ 4
4	Physic-chemical tests	AIMS 05-01-000 part 1	-	See relevant IPS
Cured				-
5	Water uptake	EN 3615, 70°C/85% R.H.	%	≤ 3

Table 9: Mechanical properties for the resin film (grade 0)

No.	Properties	Test methods	Unit	Requirements			
				RT		Dry 120°C	
				$\bar{X}$	Xmin	$\bar{X}$	Xmin
1	Tensile lap shear strength ¹⁾	AITM 1.0019 S7	MPa	²⁾	²⁾	²⁾	²⁾
¹⁾ Test not required if the resin film is the same as the one used to manufacture the corresponding prepreg material (same resin system, thickness, process and equipment...)							
²⁾ Requirements: the tests with the S7 type of samples (with film) should reach the performance level of the tests performed with the S5 type of sample (without film), both samples being manufactured with a qualified prepreg material based on the same resin as the film under qualification.							

RECORD OF REVISIONS

Issue	Clause modified	Description of modification
1 12/99		New Standard
2 01/03		Complete revision
3 06/04		Complete revision for resin film Revised information concerning the prepreg in table 2, 3, 5.1, 5.2 and 5.3
4 05/06		Line 6b in table 5.1 and 5.2 changed by MPa on GPa
5 07/13	Chapter 2 Chapter 7	AIMS05-01-000 part 100 added Chapter 7 defined in line with Incoming Inspection (Part 100) implementation